ALADDIN PORTFOLIO RISK MODEL

ANALYTICAL VAR

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Analytical VaR Methodology

Analytical VaR

Value at Risk (VaR) measures the worst expected loss under normal market conditions over a specific time interval at a given confidence level. Analytical VaR is a variance-covariance based estimate of Value at Risk. It is based on linear sensitivities of a portfolio to a broad set of systematic market risk factors and idiosyncratic risk factors. [1] In the implementation of VaR, there are two major practical problems.

- Optimize the set of risk factors to provide as much explanatory power as possible without redundancy[2]
- Calculate the risk exposure of a portfolio to these factors

BlackRock Solutions addresses these problems by maintaining and updating the set of factors used in the risk model and back-testing the model on a quarterly basis. Each security in the portfolio/benchmark is mapped to various risk factors and security-level exposures to each of these risk factors are calculated accordingly. These security-level exposures are aggregated to the compute the portfolio- and benchmark-level exposures. These portfolio- and benchmark-level exposures are then used to produce VaR/stress-testing analyses. The parametric sensitivity is calculated of every security in a portfolio to changes in key interest rates, spreads and other risk factors. The parametric exposures are summed using the appropriate portfolio weights based on different mapping methodology to compute a portfolio's exposure to these systematic market risk factors. The parametric return model uses the risk factor changes and the exposures to estimate the return of a portfolio. [3]

In Analytical VaR model, we measure the volatility σ of the portfolio returns. It is driven by systematic risk factors and idiosyncratic risk factors independently:

$$\sigma = \sqrt{\vec{e}^{\,'} \cdot \Omega \cdot \vec{e} + \vec{q}^{\,'} \cdot \Psi \cdot \vec{q}} \qquad (1)$$

where $\vec{e}$ is a vector of sensitivities of the portfolio value to changes in systematic market risk factors, Ω is the matrix of covariances between the systematic risk factors, $\vec{q}$ is a vector consisting of the market value weights of each security in the portfolio, and Ψ is the covariance matrix of the security-specific idiosyncratic risk terms. Ψ is a block diagonal matrix with the variance of the idiosyncratic return of each security in its diagonal.

In the Analytical VaR model, we assume risk factor returns are joint-normally distributed. Therefore, combinations of those risk factors, including the portfolio systematic parametric return and idiosyncratic return, will also be normally distributed. Let σ_{day} be the daily volatility of a portfolio computed by (1) using a daily covariance matrix Ω, then

$$\text{N-day } p\% \text{ AVaR} \equiv \mathrm{D}^{-1}(1 - 0.01 \cdot \mathrm{p}) \cdot \sqrt{\mathrm{N}} \cdot \sigma_{\mathrm{dav}} \qquad (2)$$

where *D()* is the cumulative distribution function of a standard normal.[4]

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[1] BRS risk model includes unexpected interest rate convexity factor as described in Appendix A.

[2] For example, for the BlackRock Fundamental Risk for Equities models, about 200 characteristics (sub-styles) were tested for explanatory power and we used what was significant in a step-wise process starting with a market and country factors. We then add industry factors giving consideration to the different levels of granularity and finally we add the style factors one-by-one. We evaluate the efficacy of these factors over the entire 15 year sample, and over three sub-samples (of 5 years) placing more significance on the more recent sub-sample of 2005-2010. We use a rule-of-thumb that any factor which is not significant ~20% of the time does not make the final cut. For fixed income spread risk models, a principal component analysis is usually used to get the key risk factors for each asset class.

[3] *Parametric Return Model*, BRS Analytics Documentation.

[4] If volatility is measured from weekly returns, then a corresponding formula may be used to estimate N-week VaR with N measured in weeks.

Covariance Matrix Estimator

Value-at-Risk analysis using the variance-covariance approach requires an estimate of the matrix of covariance between changes in risk factors. BRS provides five standard covariance calculation methodologies:

Weekly Methodologies

WKS, WKL and WKC methodologies are based entirely on weekly time series. Table 1 lists the default parameters used in these covariance estimators, WKS (weekly short-term), WKL (weekly long-term), and WKC (weekly constant weighted). Weekly estimators are more robust than daily estimators with respect to intraday lags between global market closes because a few hours lag is a relatively smaller part of a weekly period than a daily period. WKS and WKL use a short half-life exponentially weighted moving average (EWMA) to react quickly to recent changes in market volatility and correlation. Appendix A gives more technical details of the computation including the definition of overlap, an averaging technique to mitigate the impact of asynchronous data, and zero means, a method for negating the impact of trending markets.

Weekly covariance matrix takes week-over-week changes (Wednesday-to-Wednesday) of risk factors for the calculation of variance and covariance. Therefore, it gets updated on a weekly basis. Risk forecasts can only get updated once a week if exposures do not change.

Table 1: Weekly Covariance Estimator Assumptions

Methodology	Weighting	# Data Points	Overlap	Zero Means
WKS	Exponentially decay weighted Half-life = 8 Weeks, $\lambda = 0.9170$	104 Weeks	1	Yes
WKL	Exponentially decay weighted Half-life = 26 Weeks, $\lambda = 0.9737$	104 Weeks	1	Yes
WKC	Equally weighted	104 Weeks	1	Yes

Daily Methodologies

MKT and REG methodologies use daily time series for factors with daily data and weekly time series for factors with weekly data. MKT or "market weighted" methodology uses weighting with exponential decay for daily factors. REG or "regulatory weighted" uses constant weighting of the same raw time series as MKT. Table 2 summarizes assumptions used by the MKT and REG covariance estimators.

Table 2: Daily Covariance Estimator Assumptions

Factors	Methodology	Weighting	# Data Points	Overlap	Zero Means
Daily	MKT	Exponentially decay weighted, Half-life = 23 Days, $\lambda = 0.9703$	252 Days	2	Yes
	REG	Equally weighted			
Weekly	MKT,REG	Equally weighted	104 Weeks	1	Yes

Daily covariance estimators suffer from certain short-comings. Using daily time series may result in understated correlation estimates between asynchronous risk factors with time series collected at different closing times across global markets.[5]

Rolling Weekly Methodology

DLY is a rolling weekly matrix methodology. The DLY matrix updates every day instead of once a week. The rolling matrix is computed using 5-day overlapping returns that are weighted using EWMA with a 40 day half-life and 252 observations by default. The number of observation and the half-life can be specified by user. Each observation is computed as a trailing 5-day average return. Intuitively, this covariance matrix is an average of 5 different covariance matrices – one estimated with Wed-Wed data, one with Tues-Tues data, one with Mon-Mon data, etc. Each day, the covariance matrix is updated by the addition of a new rolling weekly data point for each factor in the matrix. The main advantage of the rolling matrix is that the risk forecasts are much more stable than the forecasts from the weekly matrix. The risk forecasts from the weekly matrix often jump suddenly when the matrix updates each Wednesday. The risk forecasts from the rolling matrix update more smoothly over the course of a week.

[5] A practical source of synchronous data in all sectors of global markets is not available. Even if it were, prices collected during non-business hours would be less liquid than from open markets.

Decomposition of Analytical VaR

BlackRock Solutions offers several decompositions of Analytical VaR to help isolate the contribution of individual risk factors and securities to overall portfolio risk. Those decompositions include contribution to risk and standalone risk.

- Standalone risk isolates the Analytical VaR of one or more risk factors or security exposures if all other sources of risk were hedged
- Contribution to risk identifies the change in Analytical VaR given a small percentage change in one or more risk factor or security exposures

Standalone Risk

Standard measure of risk exposure represents the security or portfolio sensitivity to changes in risk factors. But it does not say anything about the volatility of those changes. Standalone risk is a measure that isolates the risk from a particular risk factor by combining the exposure and the volatility of that factor.

Standalone risk of an individual factor is expressed as the product of risk factor volatility and risk factor exposure. For example, Table 3 reports that the test portfolio has an exposure of 0.09 years to the 3-year key rate (i.e. a 100 bps change in the 3-year key rate would change portfolio value by 0.09%). The annualized, one standard deviation volatility of the 3-year key rate is 26 basis points. Therefore, the standalone risk from the portfolio's exposure to the 3-year point is 0.09 * 26 bps/yr = 2 bps/yr.

Standalone risk of a block of factors includes correlation effects within the block, but excludes interactions between the factors in the block and other risk factors outside the block. In the example, 234 bps/yr is the portfolio standalone risk due to the total US Interest Rate exposure (key rate durations). That number excludes spread risk of the sample portfolio.

By combining exposures with volatilities, standalone risk provides information about the relative risks in a portfolio. However, because this measure ignores the correlation between risk factors or between blocks of risk factors, its information content is limited to the scenario where the particular risk factor is the only risk exposure in the portfolio (i.e. all other sources of risk have been hedged away).

Contribution to Risk and Marginal Contribution to Risk

The standalone risk measures the change in risk associated with discrete changes in factor exposure. Contribution to risk, on the other hand, is used to measure changes in risk associated with small percentage changes in factor exposure. Let us denote e_{sf} as the factor f exposure of security s and CR_{sf} as the contribution of e_{sf} to the portfolio return volatility, σ_p. Then CR_{sf} is defined to be the partial derivative of σ_p with respect to e_{sf} multiplied by e_{sf}:

$$CR_{sf} \equiv \frac{\partial \sigma_p}{\partial e_{sf}} e_{sf} \qquad (3)$$

Another measured called marginal contribution to risk captures the sensitivity of the portfolio return volatility to the factor exposure.

$$MCR_{sf} \equiv \frac{\partial \sigma_p}{\partial e_{sf}} = \frac{CR_{sf}}{e_{sf}} \qquad (4)$$

Linearity of Contribution

Additivity is a general property of the contribution to risk measure. The contribution to portfolio risk from any combination of portfolio securities and factor exposures is the sum over the individual contributions from each security specific factor exposure in the subset (i.e. the sum over CR_{sf}). This property of contribution to risk is attributable to the fact that Analytical VaR and Tracking Error are linear homogeneous functions. Linear homogeneity is the property that risk scales with the size of a portfolio. Appendix C outlines a proof that the sum of the contributions equals total portfolio risk.

Contribution to risk also differs from the other risk contribution measures in the way it is computed. The standalone contribution to risk can be calculated by zeroing out some portion of the exposure vector and re-estimating portfolio risk. In contrast, the exact value of the contribution to risk must be derived analytically since the measure corresponds to an infinitesimal change in the exposure vector rather than a finite change. Appendix C shows a quick derivation of the formula used to evaluate the contribution to risk.

A simple example helps clarify the distinction between the different decompositions. Consider a portfolio that consists of several agency bonds. This portfolio is exposed to interest rate and agency spread factors. The annualized, one standard deviation, portfolio volatility from all risk factors is 217 basis points per year. Table 3 reports the contribution to risk and standalone risk for each risk factor.

Both decompositions can be defined for an individual risk factor or security or for a block of risk factors or securities. In this example, the contribution to risk from the block of all US interest rates is 228 bps/yr. For small percentage changes in exposure, the change in portfolio volatility is equal to the contribution times the percentage change in exposure. For instance, if US interest rate exposure increased by 1% from 4.85 to 4.9, portfolio volatility would increase from 217 bps/yr to 219 bps/yr. The contribution to risk is the only risk decomposition that is additive across individual securities or factors (or blocks of securities or factors).

The standalone risk of US interest rates in this portfolio is 234bps/yr. In other words, if the portfolio could be stripped of all other risk factor exposures except US interest rates, the portfolio volatility would be 234bps/yr. Standalone risk is not additive. Notice that the sum of standalone risk figures does not equal total portfolio risk, nor does the sum of the individual US interest rate standalone figures equal the standalone risk of the whole block.

Decomposition of Analytical VaR

Table 3: Contribution to Portfolio Volatility

Risk Group	Factor Level	Factor Vol	Exposure	Type	Stand-alone Risk	Risk Contribution
Rates		matrix...	4.85		234	228
USD Int Rates		matrix...	4.85		234	228
Tsy 3Y	0.39	26	0.09	KRD	2	2
Tsy 5Y	0.80	49	4.73	KRD	231	225
Spreads		matrix...	4.87		52	-11
Other Spreads		matrix...	4.87		52	-11
Govt Related/Covered		matrix...	4.87		52	-11
US		matrix...	4.87		52	-11
Government Guarantee		matrix...	4.87		52	-11
Agency 5Y	0.16	11	4.87	Spread Duration	52	-11
Total		**matrix...**			**217**	**217**

Besides providing a useful decomposition of portfolio risk into additive components, the contribution to risk can also be used to determine the change in portfolio volatility given a small percentage change in portfolio exposure. Eq. (5) below shows that the change in portfolio volatility can be approximated by the product of the contribution to risk and the percentage change in security level factor exposure summed over all factors and all securities.

$$\Delta\sigma_p \approx \sum_f \sum_s \frac{\partial\sigma}{\partial e_{sf}}(\Delta e_{sf}) = \sum_f \sum_s CR_{sf} \frac{\Delta e_{sf}}{e_{sf}} \qquad (5)$$

Appendix A: Unexpected Convexity

The interest rate return on a fixed income security is decomposed in the BRS Risk Model into duration and convexity components. Ignoring the term structure of interest rates, the interest rate return can be represented as:

$$\frac{\Delta p}{p} \approx -OAD \cdot \Delta r + \frac{1}{2} OAC \cdot \Delta r^2 \qquad \text{(A.1)}$$

We can decompose convexity into an unexpected and expected component. We can rewrite equation (A.1) as

$$\frac{\Delta p}{p} \approx -OAD \cdot \Delta r + \frac{1}{2} OAC \cdot \left(\left(\Delta r^2 - \overline{\Delta r^2} \right) + \overline{\Delta r^2} \right) \qquad \text{(A.2)}$$

where $\left(\Delta r^2 - \overline{\Delta r^2} \right)$ and $\overline{\Delta r^2}$ represent the unexpected and expected components of convexity shock, respectively. The unexpected component is random and should be captured by the Analytical VaR model, while the expected component is deterministic and should have no impact on risk. We calibrate expected component using the square of the implied volatility from the normal ATM swaption 1M expiry contract.

Historical VaR and Stress Testing include ten convexity terms per currency. The key interest rate bucket convexity factors are

2y^2, 5y^2, 10y^2, 30y^2, 2y×5y, 2y×10y, 2y×30y, 5y×10y, 5y×30y, 10y×30y[6]

To avoid introducing too many additional risk factors into the covariance matrix, we drop the "off-diagonal" terms in the new risk model and only incorporate the four squared key rates at the 2, 5, 10, and 30yr maturities. The sum of the key rate bucket convexities is scaled to sum to option adjusted convexity. Equation A.3 shows the representation including four "unexpected" covariance shocks, and four "expected" covariance shocks. As described above, the four expected shocks are computed using the square of the normal implied volatility of the ATM swaption with 1M expiry into the four maturities (2, 5, 10, and 30yr) respectively. Only the unexpected convexity shocks are included in the covariance matrix used to compute risk. The expected convexity shocks are constructed only in an intermediate step and never show up in reports.

$$\frac{\Delta p}{p} \approx -\sum_{i=1}^{11} krd_i \cdot \Delta kr_i + \frac{1}{2} \sum_{i=1}^{4} krbc_i \cdot \underbrace{\left(\left(\Delta kr_i^2 - \sigma_{imp,i}^2 \right) \right.}_{\text{Unexpected}} \underbrace{\left. + \sigma_{imp,i}^2 \right)}_{\text{Expected}} \qquad \text{(A.3)}$$

Convexity terms for the 2, 5, 10, and 30yr maturities are included for five currencies, USD, GBP, EUR, JPY, and CAD. For the remaining currencies, there will be no granularity of convexity terms by maturity. Convexity factors are uncorrelated with all other factors in BRS Risk Model.

[6] If KRBCs are not available, 10 year convexity square factor ($10y^2$) is used

Appendix B: Estimating Covariance Matrix

The following formulae define the time series and other parameters used in estimating covariance between each pair of risk factors:

Return Series: Return of risk factor f_i for period t is defined as

$$r_{i,t} = \begin{cases} \frac{1}{K}\sum_{\tau=0}^{K-1}\left(P_{i,t-\tau} - P_{i,t-\tau-1}\right) & \text{for all yield factors and difference based spread factors} \\ -\left(\frac{\sum_{\tau=0}^{K-1} P_{i,t-\tau-1}}{\sum_{\tau=0}^{K-1} P_{i,t-\tau}} - 1\right) & \text{for all FX factors} \\ \frac{1}{K}\sum_{\tau=0}^{K-1}\left(\ln(\frac{P_{i,t-\tau}}{P_{i,t-\tau-1}})\right) & \text{for ratio based spread factors such as corporate credit factors} \\ -\frac{1}{K}\sum_{\tau=0}^{K-1}\left(\ln(\frac{P_{i,t-\tau}}{P_{i,t-\tau-1}})\right) & \text{for price based factors such as equity factors} \end{cases} \quad \text{(B.1)}$$

where K is the "overlap" parameter, and $P_{i,t}$ is the rate of factor f_i at the end of period t.

Overlap: Use of overlapping return periods as suggested by Litterman[7] minimizes the error introduced by asynchronous market closing times. A corresponding correction is made to the covariance estimate for serial correlation. BRS uses an overlap parameter K=2 for MKT and REG methodologies. This typically has a negligible effect on risk calculations for single country portfolios but tends to increase correlation across global markets.

Zero Means: BRS follows the street convention of setting the mean of all factor return time series to zero by default. Means are near zero in range-bound market environments and only become significant during trending markets precisely when a more conservative measure of risk is appropriate. Setting the mean to zero can only increase the estimate of volatility.

[7] B. Litterman, *Modern Investment Management: An Equilibrium Approach*, Wiley, 2003, chapter 16.

Appendix B: Estimating Covariance Matrix

$$\bar{r}_i = \begin{cases} \frac{1}{S(\lambda)} \cdot \sum_{t=1-n}^{0} \lambda^{-t} r_{i,t} & \text{when not setting means to zero} \\ 0 & \text{when setting means to zero} \end{cases}$$

$$\frac{1}{S(\lambda)} = \begin{cases} \frac{1-\lambda}{1-\lambda^n} & \text{for } 0 < \lambda < 1 \\ \frac{1}{n} & \text{for } \lambda = 1 \end{cases} \qquad \text{(B.2)}$$

where λ is the decay parameter and n is the number of periods used (252 days for daily and 104 weeks for weekly).

Covariance: The annualized covariance between risk factors 1 and 2 is estimated to be

$$\sigma_{1,2}^2 = m \cdot K \cdot \frac{n}{n-1} \cdot \frac{1}{S(\lambda)} \cdot \sum_{t=1-n}^{0} \lambda^{-t} (r_{1,t} - \bar{r}_1)(r_{2,t} - \bar{r}_2) \qquad \text{(B.3)}$$

where m is the number of periods in one year (252 daily or 52 weekly).

When setting the mean to zero, the simple unweighted covariance matrix is

$$\sigma_{1,2}{}^2 = m \cdot K \cdot \frac{1}{n} \sum_{t=1-n}^{0} (r_{1,t})(r_{2,t}) \qquad \text{(B.4)}$$

where m is the number of periods in one year (252 daily or 52 weekly)

Appendix C: Contribution to Risk

This appendix demonstrates that the sum of individual contributions to risk equals total portfolio risk. Equations (C.1)–(C.3) are definitions of the portfolio exposure vector, portfolio standard deviation, and the contribution to risk. Equation (C.1) defines a portfolio's exposure vector $\vec{e}_p$ as a sum over individual security exposure vectors $\vec{e}_s$ or as a double sum over e_{sf}, scalar exposures of individual securities s to single risk factors f. $\hat{e}_f$ represents a unit vector with exposure to factor f set to one and all other elements set to zero.

$$\vec{e}_p \equiv \sum_s \vec{e}_s = \sum_s \sum_f e_{sf} \hat{e}_f \qquad \text{(C.1)}$$

Equation (C.2) represents the standard deviation of return on portfolio p as the square root of the factor return covariance matrix pre- and post- multiplied by the portfolio exposure vector where e_{pf} is the exposure of the portfolio to risk factor f.

$$\sigma \equiv \sqrt{\vec{e}_p{}^T \cdot \Omega \cdot \vec{e}_p} = \sqrt{\sum_f \sum_{f'} e_{pf} e_{pf'} \sigma_{ff'}} \qquad \text{(C.2)}$$

Finally, the contribution to portfolio volatility of the security s with exposure to risk factor f is defined as the partial derivative of the portfolio standard deviation to risk factor f times the security s exposure to that factor. The second equality in equation (C.3) re-expresses the scalar contribution to risk as a dot product of two vectors which will be useful directly below.

$$CR_{sf} \equiv \frac{\partial \sigma_p}{\partial e_{sf}} \cdot e_{sf} = \frac{\partial \sigma_p}{\partial \vec{e}_p} \cdot e_{sf} \hat{e}_f \qquad \text{(C.3)}$$

The $\frac{\partial \sigma_p}{\partial \vec{e}_p}$ term in expression (C.3) is a vector of partial derivatives. Each element of the vector is a partial derivative with respect to one risk factor. This partial derivative is evaluated in equation (C.4) by expanding portfolio standard deviation and applying the chain-rule.

$$\frac{\partial \sigma}{\partial \vec{e}_p} = \frac{\partial \sqrt{\sum_f \sum_{f'} e_{pf} e_{pf'} \sigma_{ff'}}}{\partial \vec{e}_p} = \frac{\frac{\partial}{\partial \vec{e}_p}\left(\frac{1}{2}\sum_f \sum_{f'} e_{pf} e_{pf'} \sigma_{ff'}\right)}{\sigma} = \frac{\sum_f \sum_{f'} \hat{e}_f e_{pf'} \sigma_{ff'}}{\sigma} \qquad \text{(C.4)}$$

Finally, we can use equations (C.1), (C.3), and (C.4) to demonstrate that the sum of the contributions to risk over securities and factors equals the portfolio standard deviation.

$$\sum_s \sum_f CR_{sf} = \frac{\partial \sigma}{\partial \vec{e}_p} \cdot \vec{e}_p = \left[\frac{1}{\sigma}\sum_f \sum_{f'} \hat{e}_f e_{pf'} \sigma_{ff'}\right] \cdot \left[\sum_f e_{pf} \hat{e}_f\right] = \frac{1}{\sigma}\sum_f \sum_{f'} e_{pf} e_{pf'} \sigma_{ff'} = \frac{\sigma^2}{\sigma} = \sigma \qquad \text{(C.5)}$$

Appendix D: Risk Factor Summary

Asset Class	Risk Factor	Sector
Fixed Income 1,400+ Factors	Interest Rates	Government, Agency, Provincials, Inter-European Sovereign
		Interbank Spreads
	Spreads	US Agency Mortgages
		ABS, CMBS, European RMBS
		Emerging Markets
		ILBs
	Price	Non-Agency Mortgage, CMBS HY
	Volatility	Swaption
Equity 500+Factors	Stock Risk Model	Global publically listed stock universe
	World	Region, country, currency, industry, style
	EMEA	Region, country, currency, industry, style
	North America	Region, country, currency, industry, style
	Latin America	Region, country, currency, industry, style
	Japan	Country, currency, industry, style
	Asia exJapan	Region, country, currency, industry, style
FX 100+ Factors	Exchange Rates	Exchange Rates
Alternatives 250+ Factors	Hedge Fund	Hedge Fund Factors
	Commodities	1, 3 and 5 Month Futures
	Private Equity	Deal Type, Size, Geography, Industry
	Real Estate, Infrastructure	Geography, Property Type
	Leveraged Loans	Leveraged Loans